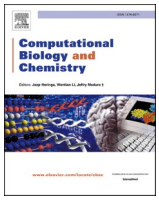
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# Multiview-cooperated graph neural network enables novel multi-omics cancer subtype classification

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ABSTRACT

Cancer presents a significant challenge in the field of public health due to its high incidence, mortality rate, and inherent heterogeneity. Integrating multi-omics biological data offers a comprehensive and intricate understanding of biological processes, disease mechanisms, and cancer subtyping, rendering it an influential tool for scientific research. Nevertheless, current approaches to integrating multi-omics data often fail to consider the scale of data in feature information and overlook the analysis of the individual and shared feature expressions in multi-omics data. The proposed study introduces a Multiview-Cooperated graph neural network (MCgnn), an end-to-end cancer subtype classifier that effectively integrates and analyzes complex multi-omics data. Firstly, MCgnn innovatively constructs a similarity network using Mahalanobis distance and density methods. Then, by employing stacked graph convolution layers, MCgnn effectively captures potential local structural features. Subsequently, MCgnn extracts and fuses complementary information from different omics data through the attention mechanism among views to achieve effective integration of multiple views and attain higher classification performance. Finally, MCgnn performs multi-task learning across omics data using a cross-omics tensor to seamlessly integrate the feature learning component with the classification component. Experiments on four publicly available The Cancer Genome Atlas (TCGA) datasets were conducted to demonstrate that MCgnn outperforms most comparison classification algorithms in effectively addressing the cancer subtype classification problem and exhibits remarkable robustness and generalization capabilities. Additionally, MCgnn was employed to identify pivotal biomarkers in cancer, providing a valuable reference for precision medicine.

## 1. Introduction

Cancer has emerged as the primary cause of mortality among individuals in both developed and developing nations worldwide, constituting a serious public health concern (Jassim et al., 2023). The formidable challenge cancer presents to clinical treatment is rooted not only in its high incidence and mortality rate but also in its remarkable heterogeneity (Manini et al., 2024). This heterogeneity is manifested by the variations among patients themselves. Even within the same patient, where tumors in different regions may exhibit distinct gene expression profiles, mutation spectra, and biological behaviors that directly impact disease diagnosis speed, treatment response, and prognosis (Tsujio et al., 2021). Consequently, relying solely on traditional approaches to determine cancer presence tends to fall short of achieving optimal treatment results for all patients (Manini et al., 2024; Tsujio et al., 2021). Therefore, the precise classification of cancer subtypes assumes pivotal importance (Jendrisak et al., 2019). Comprehensive analysis of multi-omics data allows researchers to gain profound insights into the intricate molecular characteristics of tumors (Lehar et al., 2023). This approach not only unravels new mechanisms underlying cancer development but more importantly identifies distinctive molecular markers associated with specific cancer subtypes (Ogobuiro et al., 2022). These subtypes may exhibit enhanced sensitivity to targeted therapies, thereby facilitating personalized treatment plans for patients while improving treatment success rates and minimizing unnecessary side effects (Tang et al., 2021). In summary, the precise classification of cancer subtypes represents a pivotal step towards precision medicine as it guides clinical decision-making and fosters the development of innovative drug and

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treatment strategies that instill fresh hope into the field of cancer treatment (Lehar et al., 2023).

The analysis of multi-omics information poses a central challenge in contemporary biomedical studies (Stanojevic et al., 2022). This encompasses small sample sizes, high dimensionality, significant heterogeneity, difficulties in integrating diverse technologies, and the complexity associated with interpreting biological significance (Lehar et al., 2023). These datasets encompass multiple levels including genomics, transcriptomics, proteomics, metabolomics, and more; each exhibiting unique characteristics that necessitate robust computational methodologies capable of efficient storage, processing, and integration (Miao et al., 2021). Consequently, interdisciplinary collaboration and the advancement of intelligent tools—particularly the use of AI and ML methods, have become pivotal in overcoming these challenges and advancing precision medicine (Li et al., 2025, 2024). This represents a significant stride towards gaining a deeper understanding of life sciences while enhancing human health (Xiao et al., 2023a; Zhang et al., 2023).

While various computational methods have been developed to integrate multi-omics data for cancer research, these approaches broadly encompass several categories. Beyond graph-based methods that excel at capturing complex biological relationships, other strategies include dimensionality reduction techniques combined with machine learning, such such as the PaCMAP-embedded convolutional neural network (CNN) for integrating multi-omics data into visual representations for classification (Qattous et al., 2024). Another significant direction involves generative models like Autoencoders coupled with Generative Adversarial Networks (GANs) to extract latent features and address challenges like class imbalance in multi-omics datasets (Al-Hurani et al., 2024). Despite these advancements, effectively integrating heterogeneous multi-omics data to accurately identify cancer subtypes, especially given data sparsity and inherent noise, remains a significant challenge.

Graph-based Neural Networks (GNNs) have demonstrated promising potential and efficacy in addressing multi-omics data, particularly specifically within the domain of cancer subtyping, thereby opening new avenues for advancing precision medicine (Xiao et al., 2023b; Chen et al., 2024; Liu et al., 2024). The inherent advantage of GNNs lies in their capability to directly operate on graph-structured data, which aligns perfectly with the inherent relational nature of multi-omics data. Within this framework, nodes can represent genes, proteins, or metabolites, while edges denote interactions or correlations between them, effectively capturing the intricate interplay within biological systems. This approach enables the modeling of complex networks that underpin cellular processes, providing a robust framework for uncovering disease mechanisms and identifying therapeutic targets (Han et al., 2023). For example, Graph Attention Networks (GATs) incorporate dynamic weighting of contributions from neighboring nodes, amplifying signals from key biomarkers, and achieving improved accuracy in subtyping tasks (Li and Nabavi, 2024). Similarly, Graph Convolutional Networks (GCNs) perform feature learning on the graph structures of multi-omics data by propagating and transforming feature information across nodes, effectively capturing the characteristics of local network structures (Guo et al., 2023). This facilitates the identification of specific expressions relevant to cancer subtypes, leading to superior performance in classification tasks. Recent studies have provided evidence for the robust integration of data and pattern recognition capabilities offered by GNN models in distinguishing various cancer subtypes, thus advancing precision medicine towards higher levels of sophistication (Wang et al., 2021).

The adjacency matrix is pivotal in Graph Neural Networks (GNNs) for cancer subtype classification with multi-omics data, as it directly influences information propagation between nodes. An overly sparse matrix hinders effective aggregation, while excessive density introduces noise, exacerbating the issue of over-smoothing, where models blur critical feature differences and impair classification accuracy (Li et al., 2022). Traditional adjacency matrix construction often neglects numerical feature scales, leading to a loss of biological significance and poor topology representation.

To address these challenges, researchers have explored adaptive adjacency matrices, such as weighted connectivity graphs and dynamic weight assignments, which adjust element values based on actual correlation strength, thereby enhancing biomarker sensitivity (Zhuang et al., 2023). Graph regularization techniques, like the normalized graph Laplacian (Zhu et al., 2017), mitigate over-smoothing while preserving feature diversity. Additionally, multi-layer GNN architectures with skip connections improve depth and feature retention, effectively capturing long-range dependencies (Shi et al., 2017).

However, a key limitation of existing GNN-based cancer subtyping approaches is their inadequate integration of cross-omics relationships. Many models treat genomics, transcriptomics, and proteomics as isolated data streams, employing simplistic concatenation strategies that fail to leverage inter-omics dependencies. To enhance biological insight, multimodal graph models have been proposed, integrating intra-omics structures while intentionally designing mechanisms for cross-omics interactions (Wang et al., 2022). Hypergraphs and multi-view graphs offer a more precise representation of complex regulatory networks, while multimodal fusion mechanisms allow GNNs to dynamically learn shared and distinct patterns across different omics layers (Gao et al., 2021). These innovations hold promise for more accurate and biologically meaningful cancer subtype classification, improving precision medicine by leveraging the complementary nature of multi-omics data.

Motivated by the limitations of current graph neural network methods in leveraging multi-omics data for cancer subtype classification, we propose a novel methodology called MCgnn (Multiview-Cooperated Graph Neural Network) with attention mechanisms. This method aims to fully exploit the potential biological insights embedded within multi-omics data, thereby enhancing the model's capability to comprehensively understand the complexity of diseases. MCgnn calculates the Mahalanobis distance between samples and dynamically adjusts node adjacency based on density to construct a weighted adjacency matrix. This approach not only enhances discriminability in the feature space but also effectively reduces noise and redundancy within the multi-omics data. Furthermore, MCgnn optimizes the efficient exchange of information among nodes within the network through an iterative process. Importantly, MCgnn extends the consideration of importance weights beyond individual omic datasets to the significance of single nodes across different omics datasets. This fosters information flow within each omic dataset and promotes synergistic interactions across multi-omics data. This strategy allows the model to integrate a broader range of biological features and insights, thereby improving its comprehensive understanding of the disease. After integration, MCgnn employs a multi-task learning strategy using a cross-omics tensor, enabling the seamless and comprehensive integration of diverse omics data, ultimately facilitating nuanced subtyping of samples. By harnessing these advancements, MCgnn significantly enhances the model's ability to capture the intricate and multifaceted nature of diseases, leading to more accurate and informative cancer subtype classifications.

The experimental evaluations conducted on four publicly available TCGA (Edge 2018) datasets demonstrate that MCgnn achieves high classification accuracy in cancer subtyping tasks, surpassing classical classification algorithms and most cutting-edge multi-view graph neural networks for integrating multi-omics data. This is particularly evident in challenging imbalanced multi-classification scenarios. The primary innovations of MCgnn include:

1) MCgnn is a novel architecture for constructing graph neural network structures, enhancing the aggregation of information within each class, and thoroughly incorporating the biological information concealed in the numerical scale of omics data features.

2) By employing convolution and inter-view attention mechanisms, MCgnn captures local features and facilitates the aggregation of information from different omics data.

3) MCgnn employs a multi-task learning strategy, utilizing a cross-

omics tensor to comprehensively and multidimensionally integrate diverse omics information, improving the model's ability to generalize, optimize data utilization, and accelerate learning.

The rest of this paper is structured as follows. Section 2 provides a summary of relevant background information. In Section 3, the framework of the proposed method is comprehensively presented and discussed. Section 4 is dedicated to experimentally evaluating the proposed method using four TCGA datasets, and comparing it with other competing algorithms. Lastly, Section 5 concludes and outlines future research directions for this study.

## 2. Related work

The proposed methodology incorporates Multi-View Graph Neural Networks (MV-GNNs) (Wang et al., 2021), Graph Convolutional Networks (GCNs) (Guo et al., 2023), and Attention Mechanisms (Li and Nabavi, 2024). This section presents a summary of the prior studies conducted on these three components.

### 2.1. Multi-view graph neural networks (MV-GNNs)

Multi-view learning is an approach that trains models on data from multiple sources (Chen et al., 2024; Ouyang et al., 2023; Liang et al., 2024). It aims to synthesize consistent and complementary insights from these varied sources while reducing the impact of redundant data. In real-world applications, a single entity can be viewed from multiple angles, resulting in multiple perspectives on the same object. Different perspectives often contain unique details that can be leveraged to enrich each other, thereby providing a more holistic understanding of the entity.

The deep exploration of coherent and supplementary knowledge enables a superior acquisition of intrinsic representations for multi-view data (Lan et al., 2024; Liu et al., 2024). For example, A novel approach was introduced by Nie et al (Nie et al., 2016)., where they reconstructed the conventional spectral learning model to enable adaptive weight learning for each view. Additionally, they explored a token matrix through the aggregation of loss functions from multiple perspectives. In their work, Jiang and Liu et al (Jiang and Liu, 2022). devised an innovative technique for merging similarity relationships acquired from diverse viewpoints to achieve comprehensive topological representation. They also incorporated a trainable weight parameter to effectively combine valuable topological information. Both approaches emphasize capturing consistent information from diverse perspectives using different constraints aimed at learning effective representations. In addition, MV-GNNs have demonstrated remarkable proficiency in tackling the challenge posed by multi-omics data, effectively capturing its complexities and heterogeneities while uncovering intrinsic connections among data points (Wang et al., 2021; Li et al., 2022).

### 2.2. Graph Convolution

Graph Convolution plays a pivotal role in the analysis of multi-omics information. This operation facilitates the aggregation of local information by leveraging interactions between node features and the adjacency matrix. This allows the network to capture the comprehensive characteristics of individual nodes and their neighbors. Consequently, it enables learning a more intricate and nuanced feature representation, which is essential for discerning complex patterns within biological datasets (Wang et al., 2021). This characteristic makes it highly suitable for various research domains, such as Wang et al.'s adjustable multi-channel graph convolutional network that utilizes an attention mechanism to extract representations from different aspects of the nodes (Wang et al., 2020). Additionally, Feng et al (Feng et al., 2021). introduced a novel cross-feature graph convolution operator with linear complexity to the feature dimension. Ultimately, these approaches aim to achieve high performance even with limited labeled data by leveraging the power of GCN in representation learning and aggregation tasks across diverse research areas.

### 2.3. Attention mechanism

In contrast to convolution-based weight calculation, the attention-based method has gained significant traction in recent years. For example, Veličković et al (Veličković et al., 2017). proposed a GNN framework that incorporates a self-attentive mechanism to assign different weight coefficients to neighboring nodes without considering the graph construction. Chen et al (Chen et al., 2021). extended the application of the graph attention network by simultaneously incorporating node and edge features in iterative processes. Graph Attention Mechanism (GAM) enhances the discrimination of feature correlations across distinct omics layers, thereby improving the model's comprehension of complex interrelationships. Remarkably, GAM enhances model interpretability while ensuring computational efficiency even when dealing with extensive node sets and intricate relational structures.

## 3. Proposed methodology

In this section, we propose a novel cancer subtype classification algorithm for microarray datasets called MCgnn (Multiview-Cooperated graph neural network enables novel multi-omics cancer subtype classification). We start with the overview framework for MCgnn (Section 3.1). Then, we present the three main components of our algorithms: construction of graph neural network (Section 3.2), extraction and fusion of multi-omics data features (Section 3.3) and loss function of multi-task learning (Section 3.4).

### 3.1. Overview framework of MCgnn

A concise summary of MCgnn process is illustrated in Fig. 1 MCgnn constructs a MV-GNN structure for preprocessed multi-omics datasets, where each view corresponds to a distinct omics layer. To quantify sample similarity, MCgnn employs covariance matrices and their inverses to compute the Mahalanobis distance. The proposed method dynamically adjusts node adjacency based on density (as explained in **Section 3.2**) to construct a weighted adjacency matrix for multi-omics data. Within this framework, each view represents specific omics data with nodes representing samples, and each node has replicas across other views that capture its feature expression across different omics data layers. The edges connecting nodes signify relationships and correlations within each view, reflecting inherent connections among samples within the context of that particular omics layer.

Detailed explanations of view operations are provided in **Section 3.3**. Initially, the constructed Graph Neural Network undergoes multiple layers of convolution to gradually extract higher-level abstractions of node features, thereby playing a pivotal role in uncovering underlying biological principles and significant feature combinations. Subsequently, an adaptive parameter is introduced to evaluate the importance weights of various omics datasets, extending this assessment to the significance of various omics information for the same sample. This leads to the implementation of inter-view attention mechanisms for samples, facilitating information exchange within individual omics layers and fostering collaborative interactions across omics data to promote cross-omics data integration. By employing this approach, MCgnn effectively ensures that the model harnesses synergistic advantages from various omics layers comprehensively depicting biological phenomena, thus enhancing both accuracy and interpretability in cancer subtype classification.

In **Section 3.4**, the details of integrating multi-task learning within the multi-view framework for classification tasks are discussed extensively. After information fusion, MCgnn extracts prediction labels from each view and converts them into vectors. By leveraging the concept of a

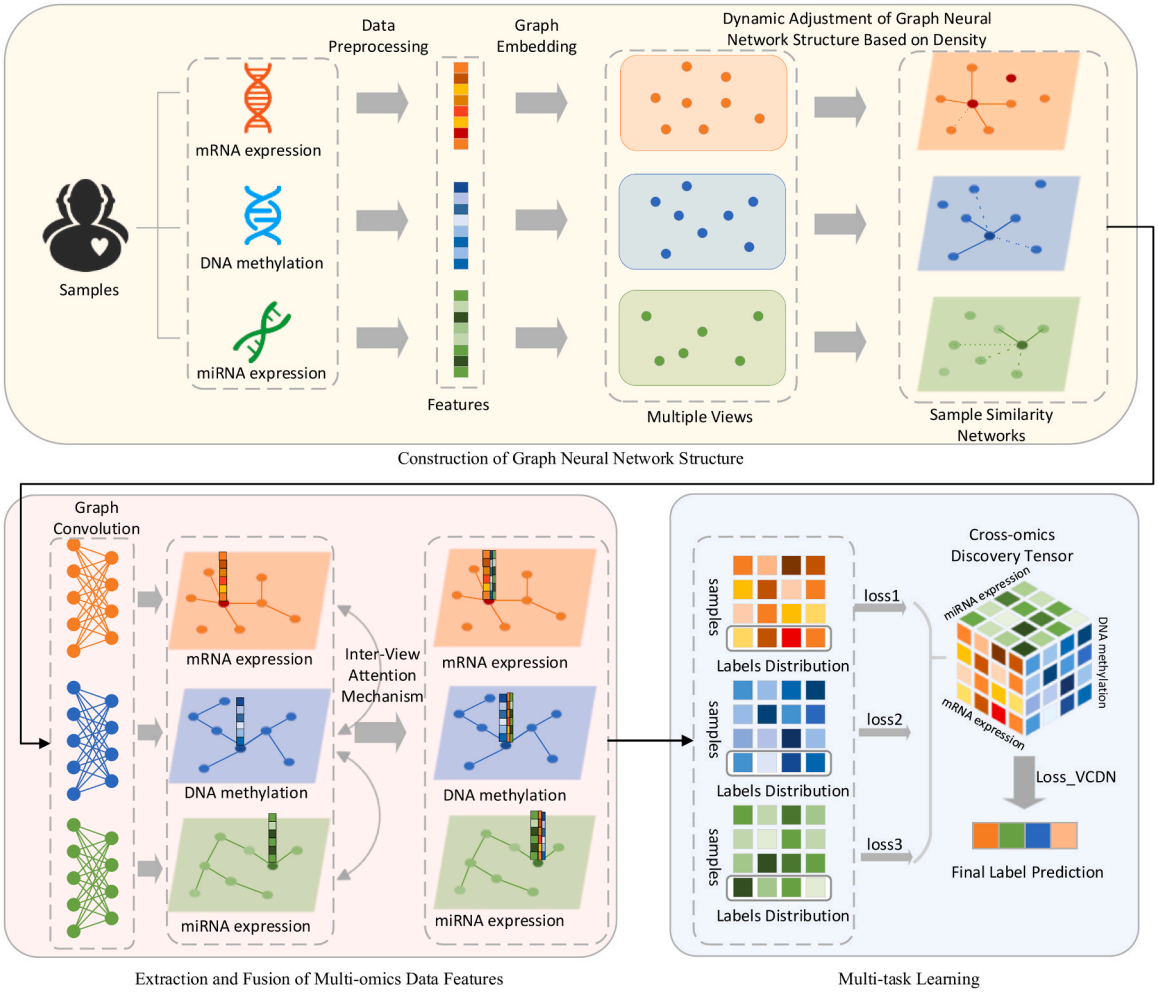



**Fig. 1.** Schematic diagram of MCgnn.

cross-omics tensor, it generates a multi-dimensional tensor that synthesizes all available information to derive the final prediction. The construction of a multi-task learning strategy involves optimizing an objective function that holistically integrates diverse omics data, enabling fine-grained subtyping of samples based on combined insights from multiple omics layers. The optimization function aims to minimize overall error across all tasks while promoting shared patterns and features between tasks. This approach not only enhances generalization capabilities across different omics types but also improves prediction robustness and accuracy by harnessing the complementary nature of the data.

### 3.2. Construction of graph neural network structure

Given the complexity of handling multi-omics datasets, each characterized by distinct feature representations and complementary information, it becomes imperative to construct separate Graph Neural Networks (GNNs) for each omics dataset. Initially, each omics dataset is transformed into a manipulable data structure: a data matrix $\boldsymbol{X} \in \boldsymbol{R}^{\boldsymbol{N}*\boldsymbol{M}}$, where $\boldsymbol{N}$ represents the total number of samples and $\boldsymbol{M}$ denotes the total quantity of features. Assuming simultaneous processing $\boldsymbol{n}$ omics datasets, the input multi-omics datasets are converted into $\boldsymbol{n}$ corresponding matrices $X_1, X_2, \ldots, X_n$.

Subsequently, adjacency matrices are constructed to investigate the similarity among samples, facilitating the establishment of a similarity network that captures the local structural information between samples. In this study, an innovative approach is proposed by utilizing the Mahalanobis distance as a metric between samples and constructing the network topology based on density. By dynamically adjusting node adjacency, informational noise and redundancy are efficiently mitigated, thereby achieving high-quality information transfer among nodes.

For a given data matrix $\boldsymbol{X}$, the calculation of the covariance matrix $\boldsymbol{\Sigma}$ is as follows:

$$\Sigma = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})(x_i - \overline{x})^T + \varepsilon I\# \tag{1}$$

Where, $\boldsymbol{x_i}$ represents the $\boldsymbol{i}$-th sample of the data, and $\overline{\boldsymbol{x}}$ denotes the mean values of the sample features. The regularization parameter $\boldsymbol{\varepsilon}$ is introduced to prevent singularity in the covariance matrix when dealing with a high-dimensional dataset (e.g.,$\boldsymbol{M}$>$\boldsymbol{N}$). $\boldsymbol{I}$ represents an identity matrix used for regularization purposes. Covariance serves as a statistical measure that quantifies the correlations between variables within a multidimensional dataset. When analyzing more than two variables, a covariance matrix is utilized to assess interrelationships among multiple variables. The mean values $\overline{\boldsymbol{x}}$ is calculated as $\overline{x}_{,a} = \frac{1}{n}\sum_{i=1}^{N} x_{i,a}$, where $\overline{\boldsymbol{x}}_{,\boldsymbol{a}}$ denotes the average value of the $\boldsymbol{a}$-th feature across all samples, and $\boldsymbol{x}_{\boldsymbol{i},\boldsymbol{a}}$ denotes the value of the $\boldsymbol{a}$-th feature for the $\boldsymbol{i}$-th sample. Subsequently, the Mahalanobis distance between samples is calculated:

$$d_{ij} = \sqrt{(x_i - x_j)^T \Sigma^{-1}(x_i - x_j)}\# \tag{2}$$

Where, $\boldsymbol{\Sigma^{-1}}$ denotes the inverse matrix of $\boldsymbol{\Sigma}$, and $\boldsymbol{d_{ij}}$ denotes the Maha-

lanobis distance between the ***i***-th and ***j***-th samples. The utilization of Mahalanobis distance in the proposed methodology is motivated by its ability to standardize distances using the covariance matrix, effectively accounting for correlations among features. Moreover, Mahalanobis distance normalizes the scales of different features to prevent any single feature from dominating the distance calculation. This characteristic proves particularly advantageous when dealing with high-dimensional data and features exhibiting inherent correlations. The superiority of Mahalanobis distance in representing the topological distribution among sample points, compared to Euclidean distance, becomes evident when variations in data scale normalization are considered (Fig. 2). A comprehensive consideration of nodes' topological distribution facilitates the design of more effective graph neural network models, thereby yielding improved results across various applications.

$d_{ij}$ are placed into a $N \times N$ Mahalanobis distance matrix $D$, and then the density $\rho_i$ of each sample point is calculated as follows:

$$\rho_i = \frac{1}{k} \sum_{j \in N_k(i)} d_{ij} \# \tag{3}$$

Where $N_k(i)$ denotes the $k$-nearest neighbor set of the $i$-th sample. The parameter $k$, which serves as a user-defined hyperparameter, is utilized as a scaling factor to control the threshold. Based on this, the weighted adjacency matrix $A$ is constructed based on the subsequent guideline:

$$A_{ij} = \begin{cases} \frac{1}{1+d_{ij}}, \& if d_{ij} \le \rho_i \bullet k \\ 0 \quad , \& otherwise \end{cases} \# \tag{4}$$

The formula (4) implies the edges between nodes with distances below the density threshold are preserved, which significantly facilitates the graph neural network in learning sample features and distinguishing categories. Each node dynamically establishes connections with its neighbors, promoting information aggregation and effectively filtering out noise.

The density-weighted adjacency matrices are computed and normalized for both training and testing data sets, thereby enhancing the graph learning model's comprehension of data structures, reducing noise interference, and optimizing training efficiency as well as model performance. The normalized Laplacian matrix $L$ is calculated using the following formula:

$$D = diag\left( \sum_j A_{ij} \right) \# \tag{5}$$

$$D^{-\frac{1}{2}} = diag\left( \left( \sum_j A_{ij} \right)^{-\frac{1}{2}} \right) \# \tag{6}$$

$$L = I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \# \tag{7}$$

Where $D$ is the degree matrix, $D^{-1/2}$ is the inverse square root of the degree matrix, and $I$ is the identity matrix. The normalized Laplacian matrix effectively balances information propagation between nodes, mitigates the impact of degree distribution imbalance, alleviates the over-smoothing problem, and enhances the capability of graph neural networks to capture both local and global structural information.

### 3.3. Extraction and fusion of multi-omics data features

After constructing the complete MV-GNN, multi-omics data features are extracted and fused by stacking graph convolutional layers and inter-view attention layers. Given an input feature matrix $X$ and adjacency matrix $L$, the output of the graph convolution layer can be expressed as:

$$H = \sigma(LXW + b) \# \tag{8}$$

Where $X \in R^{N \times F}$ is the input feature matrix, $N$ is the number of nodes, and $F$ is the dimension of input features. $L \in R^{N \times N}$ is the adjacency matrix. $W \in R^{F \times F'}$ is a trainable weight matrix, where $F'$ is the dimension of output features, and $b \in R^{F'}$ is the bias vector. $\sigma$ denotes the activation function. Multiple graph convolutional layers can be stacked to explore deeper feature information.

The inter-view attention mechanism facilitates information exchange across various omics datasets by calculating a weighted feature combination for each view. Given the feature representations $H_1, H_2, \ldots, H_n$ from multiple views, a linear transformation is applied to the feature representation of each view. For the $v$-th view, the feature representation after linear transformation is:

$$H'_v = H_v W_v \# \tag{9}$$

where $W_v$ is the linear transformation matrix for the $v$-th view. Subsequently, for each view, its attention score is computed as follows:

$$score_v = \sum_{v=1}^{V} H'_v \odot \beta_v \# \tag{10}$$

where $\beta_v$ represents attention weight vector for the $v$-th view, which is a trainable parameter of the model updated through backpropagation

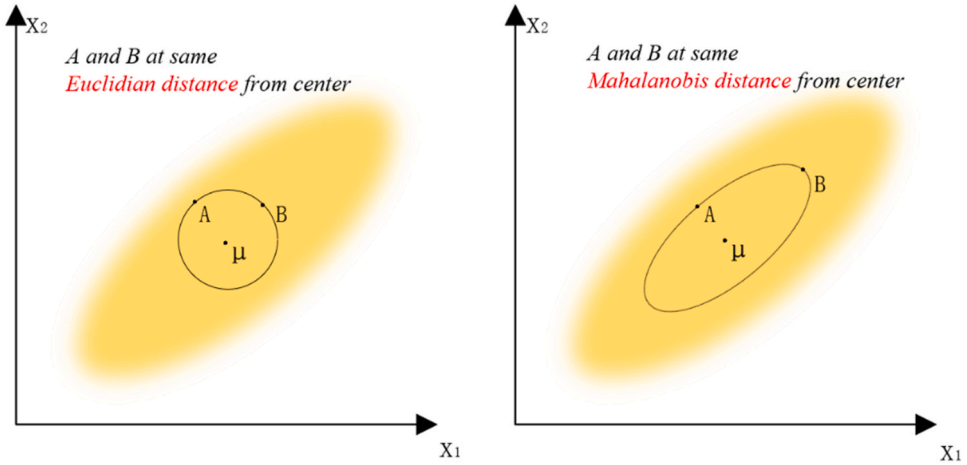



**Fig. 2.** Schematic diagram of Euclidean distance and Mahalanobis distance, for node topological distribution.

during training. The operator ⊙ denotes the dot product, resulting in $score_v$ as a scalar value. To obtain the attention weights, these scores are normalized by applying the softmax function:

$$\alpha_v = \frac{\exp(score_v)}{\sum_{v'}\exp(score_{v'})}\# \tag{11}$$

The weighted feature combination is represented as:

$$H^{att} = \sum_{v=1}^{V} \alpha_v H_v'\# \tag{12}$$

$\boldsymbol{H^{att}}$ represent the aggregated feature expressions transferred among all views. For the $\boldsymbol{v}$-th view, its original feature expression is retained, and the weighted sum of feature expressions transferred from all other views is added to compute the final feature representation as follows:

$$H_v^{final} = H_v + H^{att} - \alpha_v H_v'\# \tag{13}$$

### 3.4. Loss function of multi-task learning

The proposed MCgnn conducts independent classification for the feature expressions of each view and employs the View Correlation Discovery Network (VCDN) for joint classification of features from all views. Independent classification is conducted for the features of each view $\boldsymbol{v}$. The loss function employed is the cross-entropy loss function, which is calculated as follows:

$$\mathscr{L}_{ind}^{v} = \frac{1}{N}\sum_{i=1}^{N} CrossEntropy\left(y_i, \hat{y}_i^{v}\right)\# \tag{14}$$

where $\boldsymbol{y_i}$ denotes the real label of the $\boldsymbol{i}$-th sample, and $\hat{\boldsymbol{y}}_{\boldsymbol{i}}^{\boldsymbol{v}}$ denotes the prediction result of the classifier for the $\boldsymbol{i}$-th sample from view $\boldsymbol{v}$. ***CrossEntropy()*** refers to the cross-entropy loss function utilized to measure the discrepancy between the true labels and the predicted probabilities.

The joint classifier VCDN is responsible for the joint classification of features from multiple views. Initially, for each sample $\boldsymbol{j}$, the predicted label distributions from all omics data, $\hat{\boldsymbol{y}}_{\boldsymbol{j}}^{(\boldsymbol{v})}$,$\boldsymbol{v}$= ***1, 2, …, n***, serve to construct a cross-omics discovery tensor $\boldsymbol{C_j} = \hat{\boldsymbol{y}}_{\boldsymbol{j}}^{(\boldsymbol{1})}\hat{\boldsymbol{y}}_{\boldsymbol{j}}^{(\boldsymbol{2})}\ldots\hat{\boldsymbol{y}}_{\boldsymbol{j}}^{(\boldsymbol{n})} \in \boldsymbol{R^{N^n}}$. This tensor $\boldsymbol{C_j}$ is then passed through the ***VCDN()*** for final prediction. ***VCDN()*** is structured as a fully connected network with an output dimension of $\boldsymbol{N}$, where $\boldsymbol{N}$ denotes the number of samples, utilizing a cross-entropy loss function, which is computed as follows:

$$\mathscr{L}_{VCDN} = \frac{1}{N}\sum_{i=1}^{N} CrossEntropy\left(y_i, \hat{y}_i^{VCDN}\right)\# \tag{15}$$

where $\hat{\boldsymbol{y}}_{\boldsymbol{i}}^{\boldsymbol{VCDN}}$ is the joint prediction result of VCDN for the $\boldsymbol{i}$-th sample. The aggregate loss function for multitask learning encompasses the sum of the independent classification losses from all views and the joint classification loss. This approach considers both the individual predictions made from each omics dataset and the collective predictions derived from the amalgamation of all omics datasets. The specific expression is as follows:

$$\mathscr{L}_{total} = \sum_{v=1}^{n}\mathscr{L}_{ind}^{v} + \mathscr{L}_{VCDN}\# \tag{16}$$

## 4. Experimental results and analysis

This section evaluated the efficacy of the MCgnn model through a comparative analysis with established classification algorithms designed for multi-omics datasets. To impartially evaluate the integration methods across various multi-omics datasets, five-fold cross-validation process was implemented on four datasets. Additionally, ablation experiments were conducted to confirm the robustness and stability of MCgnn. Furthermore, MCgnn was utilized to identify significant biomarkers in cancer, providing a valuable insight for precision medicine.

### 4.1. Dataset and evaluation metrics

To demonstrate the efficacy of the proposed methodology, MCgnn was employed in four distinct biomedical classification tasks: distinguishing Alzheimer's Disease (AD) patients from normal controls (NC) using the ROSMAP dataset, grading low-grade glioma (LGG) in the LGG dataset, classifying renal cancer types in the KIPAN dataset, and identifying breast invasive carcinoma (BRCA) PAM50 subtypes in the BRCA dataset. These tasks involved leveraging three different omics data types: mRNA expression data (mRNA), DNA methylation data (DNA methylation), and miRNA expression data (miRNA). These datasets offer comprehensive and complementary insights for disease characterization. To mitigate any potential impact of missing or redundant features on classification performance, preprocessing and feature selection were conducted independently for each omics data type. Further details regarding these datasets can be found in Table 1.

Omics datasets for LGG, KIPAN, and BRCA, along with grading information for LGG patients, were obtained through the Broad GDAC Firehose from The Cancer Genome Atlas (TCGA) program. PAM50 is a 50-gene signature that classifies breast cancer into five molecular subtypes: normal-like, basal-like, HER2-enriched, Luminal A, and Luminal B (Parker et al., 2009; Brigham et al., 2012). Information on the PAM50 breast cancer subtypes for TCGA BRCA patients was obtained through TCGABiolinks (Colaprico et al., 2016). The LGG dataset involves the grading classification of low-grade gliomas (LGG). For the classification of renal cancer types, renal cancer is categorized into three distinct subtypes: chromophobe renal cell carcinoma (KICH), clear cell renal cell carcinoma (KIRC), and papillary renal cell carcinoma (KIRP). Different omics data in the ROSMAP dataset are sourced from the AMP-AD Knowledge Portal. ROSMAP comprises two longitudinal clinical-pathological cohort studies of Alzheimer's Disease (AD) from Rush University: ROS and MAP.

The performance of MCgnn in classifying multi-omics data was compared to that of existing classification algorithms. Additionally, extensive ablation studies were conducted to highlight the significance of various components within MCgnn. The algorithms were evaluated using key metrics such as accuracy (Acc), weighted F1 score (F1_weighted), and balanced accuracy (BACC).

In classification tasks, a model's prediction outcomes can be summarized using four key indicators: TP (true positives), FP (false positives), FN (false negatives), and TN (true negatives). Accuracy is the proportion of correctly classified samples out of all samples. Accuracy (Acc), calculated as the proportion of correctly classified samples by the classifier out of all samples, provides an overall measure of prediction correctness. However, in multi-class problems, especially those with imbalanced class distributions, accuracy alone can be misleading as it may be disproportionately influenced by the performance on the majority classes. It is calculated as follows:

$$Acc = \frac{TP + TN}{TP + FP + TN + FN}\# \tag{17}$$

**Table 1**
Statistics of dataset features and labels.

| Datasets | Number of omics data features | | | Categories |
|---|---|---|---|---|
| | mRNA | DNAmethy | miRNA | |
| BRCA | 1000 | 1000 | 503 | Normal-like: 115 , Basal-like: 131 , HER2-enriched: 46, Luminal A: 436 , Luminal B: 147 |
| LGG | 2000 | 2000 | 548 | Grade 2: 246 , Grade 3: 264 |
| KIPAN | 2000 | 2000 | 445 | KICH: 66 , KIRC: 318 , KIRP: 274 |
| ROSMAP | 200 | 200 | 200 | NC: 169 , AD: 182 |

The F1 score is the harmonic mean of precision and recall, incorporating both misclassification and missed classifications by the model. To better account for imbalances in sample distribution, the weighted F1 score provides an enhanced evaluation of minority classes. By assigning higher weights to minority classes, the weighted F1 score encourages improved recognition capabilities specifically for them, thereby achieving better class balance in multi-class problems. The formal definition is as follows:

$$F1score = 2 \times \frac{pression \times recall}{pression + recall} \# \tag{18}$$

$$F1_{weighted} = \frac{\sum_{i=1}^{n} \omega_i \times F1_i}{\sum_{i=1}^{n} \omega_i} \# \tag{19}$$

Here, $\boldsymbol{precision}$ is defined as $\boldsymbol{pression} = \frac{TP}{TP+FP}$, and $\boldsymbol{recall}$ is defined as $\boldsymbol{recall} = \frac{TP}{TP+FN}$, where $\boldsymbol{n}$ represents the total number of classes, $\boldsymbol{F1}_i$ represents the F1 score for the $\boldsymbol{i}$-th class, and $\boldsymbol{\omega}_i$ represents the weight for the $\boldsymbol{i}$-th class, which is the proportion of samples in that class relative to the total number of samples.

Balanced accuracy (BACC) is the macro-average of the accuracies of all classes, considering the predictive performance across all classes, rather than focusing predominantly on a specific class. For each class, the balanced accuracy calculates the accuracy of that class and then averages these values, ensuring that even in cases of unbalanced class distribution, the model's performance is assessed more equitably. Balanced accuracy (BACC) is the macro-average of the accuracies of all classes. This metric explicitly addresses challenges posed by imbalanced class distributions by ensuring that the predictive performance across all classes is considered equally, rather than being dominated by larger classes. For cancer subtype classification, where certain subtypes may be rarer, BACC provides a more robust and fair assessment of the model's ability to classify across all distinct cancer subtypes, offering a more nuanced understanding than overall accuracy. It is defined as follows:

$$BACC = \frac{\sum_{i=1}^{n} acc_i}{n} \# \tag{20}$$

where $\boldsymbol{acc}_i$ is the accuracy for the $\boldsymbol{i}$-th class and $\boldsymbol{n}$ denotes the number of classes. All methods were evaluated using five-fold cross-validation experiments and the average values and standard deviations of the evaluation metrics were subsequently compiled and reported. The test set data were not involved in the training process and were used solely to measure the performance of the algorithms in integrating multi-omics data and their generalizability. Additionally, all experiments were performed under identical hardware and software conditions.

### 4.2. Model parameter configuration

The comprehensive configuration of hyperparameters, regularization strategies, optimizers, and activation functions for the MCgnn model is detailed in this section. These settings were meticulously chosen to ensure optimal performance and robustness across diverse multi-omics datasets.

The GCN encoder (GCN_E) is structured with three convolutional layers, utilizing hidden layer dimensions of [128,256,256] as defined. To mitigate overfitting, a dropout rate of 0.5 was consistently applied after each convolutional layer within GCN_E. In the initial pre-training phase, the model underwent 100 epochs, with a learning rate of 1e-3 for both the encoders and the view-specific classifiers. Following pre-training, the main training phase extended for 200 epochs. During this phase, the learning rate for the encoders and view-specific classifiers was set to 5e-4, while the learning rate for the VCDN classifier was 1e-3.

A critical hyperparameter, the number of nearest neighbors (k) considered for constructing the sample similarity network based on density, the selection and impact of this numerical setting are further elaborated and discussed in detail in Section 4.6 **Hyperparameter Evaluation**. This parameter plays a crucial role in balancing the sparsity and density of the adjacency matrix, thereby influencing information aggregation. The hidden layer dimension for the VCDN classifier was set to 256, aligning with the output dimension of the GCN encoder to ensure consistent feature representation for multi-task learning.

For regularization, a minor term of 1e-10 was added to the covariance matrix during the Mahalanobis distance calculation to prevent singularity when handling high-dimensional datasets. All model parameters were optimized using the Adam optimizer. Regarding activation functions, LeakyReLU with a negative slope of 0.25 was employed after the convolutional layers in GCN_E and within the VCDN model. Prior to concatenation in VCDN, the flattened features from each view were processed by a Sigmoid activation function. The Softmax function was utilized for normalizing attention scores within the inter-view attention mechanism and for generating the final probability distributions in the classification output.

### 4.3. Compared with related algorithms

The performance of MCgnn in classifying data was evaluated by comparing it with five conventional machine learning classification algorithms, three deep learning techniques, and three MV-GNN-based multi-omics data classification algorithms: (1) k-Nearest Neighbors (KNN): An instance-based learning algorithm that classifies new instances by determining the category of the most similar training samples based on distance metrics. (2) Support Vector Machine (SVM): a supervised learning model that uses a hyperplane to separate data points in a way that maximizes the margin between different classes. (3) Random Forest (RF): A technique for enhancing classification accuracy by creating numerous decision trees and combining their forecasts, thereby boosting the performance of ensemble learning. (4) Logistic Regression: A probability-based linear classification method that transforms the output of linear regression into probability values to predict the likelihood of an event and classify it accordingly. (5) Naive Bayes (NB): a simple probabilistic classifier based on applying Bayes' theorem with feature independence assumptions. (6) Multi-layer Perceptron (MLP): A neural network based on fully connected layers, suitable for various classification tasks. (7) Generative Adversarial Networks (GANs): Enhances data quality and classification through adversarial training between a generator and a discriminator, used for data augmentation. (8) Transformer: Processes sequence data through self-attention mechanisms, which are applied to omics data analysis for classification tasks. (9) MOGCN (Li et al., 2022): An integrative method for multi-omics data based on graph convolutional networks, which builds a patient similarity network through autoencoder dimensionality reduction and similarity network fusion for cancer subtype classification and clinical diagnosis. (10) MOGONET (Wang et al., 2021): Integrates multi-omics data using graph convolutional networks for patient classification and biomarker identification. (11) CustOmics (Benkirane et al., 2023): A versatile deep-learning based strategy for multi-omics integration. (12) AttentionMOI (Pang et al., 2023): A denoised multi-omics integration framework for cancer subtype classification and survival prediction. (13) MOGAT (Tanvir et al., 2024): A multi-omics integration framework utilizing multi-head attention mechanisms for cancer subtype prediction. The preprocessed data were uniformly utilized for training all techniques. The classification outcomes for the BRCA, LGG, KIPAN, and ROSMAP datasets are displayed in Table 2 and Table 3.

Table 2 and Table 3 show that MCgnn outperforms most of the compared algorithms in multi-omics data classification tasks, with only a slight disadvantage relative to Naive Bayes on the BRCA dataset in terms of balanced accuracy (BACC). In addition, MCgnn exhibits minimal experimental variance particularly on the BRCA dataset. These findings highlight MCgnn's exceptional capabilities in multi-omics data

**Table 2**
Comparison of MCgnn and other classification models on BRCA and LGG datasets.

| Methodology | BRCA | | | LGG | | |
|---|---|---|---|---|---|---|
| | Acc | F1_weighted | BACC | Acc | F1_weighted | BACC |
| KNN | 0.718 ± 0.034 | 0.705 ± 0.048 | 0.628 ± 0.052 | 0.694 ± 0.038 | 0.693 ± 0.039 | 0.696 ± 0.037 |
| SVM | 0.714 ± 0.019 | 0.719 ± 0.014 | 0.677 ± 0.042 | 0.731 ± 0.038 | 0.730 ± 0.039 | 0.735 ± 0.041 |
| RF | 0.733 ± 0.022 | 0.725 ± 0.020 | 0.619 ± 0.027 | 0.715 ± 0.045 | 0.714 ± 0.045 | 0.719 ± 0.046 |
| Logistic Regression | 0.715 ± 0.014 | 0.707 ± 0.039 | 0.650 ± 0.034 | 0.690 ± 0.028 | 0.689 ± 0.029 | 0.690 ± 0.028 |
| NB | 0.725 ± 0.047 | 0.743 ± 0.036 | **0.776 ± 0.046** | 0.668 ± 0.038 | 0.651 ± 0.039 | 0.677 ± 0.046 |
| MLP | 0.759 ± 0.014 | 0.760 ± 0.031 | 0.711 ± 0.025 | 0.721 ± 0.022 | 0.719 ± 0.024 | 0.722 ± 0.023 |
| GANs | 0.762 ± 0.018 | 0.758 ± 0.044 | 0.719 ± 0.032 | 0.715 ± 0.025 | 0.714 ± 0.026 | 0.715 ± 0.026 |
| Transfomer | 0.764 ± 0.017 | 0.767 ± 0.029 | 0.716 ± 0.019 | 0.729 ± 0.025 | 0.729 ± 0.025 | 0.730 ± 0.026 |
| Mogonet | 0.787 ± 0.029 | 0.775 ± 0.030 | 0.674 ± 0.051 | 0.733 ± 0.085 | 0.729 ± 0.089 | 0.732 ± 0.084 |
| Mogcn | 0.777 ± 0.022 | 0.764 ± 0.019 | 0.670 ± 0.029 | 0.712 ± 0.045 | 0.710 ± 0.033 | 0.714 ± 0.044 |
| CustOmics | 0.782 ± 0.013 | 0.790 ± 0.011 | 0.764 ± 0.029 | 0.704 ± 0.023 | 0.696 ± 0.023 | 0.705 ± 0.021 |
| AttentionMOI | 0.745 ± 0.159 | 0.750 ± 0.165 | 0.752 ± 0.102 | 0.700 ± 0.037 | 0.692 ± 0.041 | 0.693 ± 0.033 |
| MOGAT | 0.777 ± 0.022 | 0.759 ± 0.015 | 0.668 ± 0.039 | 0.706 ± 0.035 | 0.706 ± 0.050 | 0.710 ± 0.033 |
| MCgnn | **0.791 ± 0.021** | **0.791 ± 0.008** | 0.742 ± 0.012 | **0.771 ± 0.025** | **0.769 ± 0.026** | **0.770 ± 0.028** |

**Table 3**
Comparison of MCgnn and other classification models on KIPAN and ROSMAP datasets.

| Methodology | KIPAN | | | ROSMAP | | |
|---|---|---|---|---|---|---|
| | Acc | F1_weighted | BACC | Acc | F1_weighted | BACC |
| KNN | 0.938 ± 0.023 | 0.938 ± 0.023 | 0.922 ± 0.031 | 0.666 ± 0.060 | 0.666 ± 0.061 | 0.668 ± 0.060 |
| SVM | 0.953 ± 0.019 | 0.953 ± 0.019 | 0.933 ± 0.029 | 0.749 ± 0.043 | 0.748 ± 0.044 | 0.750 ± 0.042 |
| RF | 0.956 ± 0.021 | 0.956 ± 0.020 | 0.939 ± 0.035 | 0.709 ± 0.046 | 0.708 ± 0.065 | 0.711 ± 0.064 |
| Logistic Regression | 0.964 ± 0.012 | 0.964 ± 0.012 | 0.957 ± 0.016 | 0.777 ± 0.047 | 0.777 ± 0.048 | 0.776 ± 0.048 |
| NB | 0.903 ± 0.035 | 0.902 ± 0.035 | 0.906 ± 0.040 | 0.734 ± 0.038 | 0.732 ± 0.040 | 0.735 ± 0.036 |
| MLP | 0.954 ± 0.021 | 0.955 ± 0.021 | 0.947 ± 0.027 | 0.769 ± 0.058 | 0.768 ± 0.058 | 0.765 ± 0.056 |
| GANs | 0.951 ± 0.023 | 0.952 ± 0.023 | 0.952 ± 0.027 | 0.803 ± 0.042 | 0.803 ± 0.042 | 0.804 ± 0.041 |
| Transfomer | 0.965 ± 0.014 | 0.965 ± 0.014 | 0.955 ± 0.023 | 0.800 ± 0.047 | 0.799 ± 0.046 | 0.798 ± 0.043 |
| Mogonet | 0.971 ± 0.024 | 0.971 ± 0.024 | 0.960 ± 0.018 | 0.826 ± 0.026 | 0.827 ± 0.026 | 0.829 ± 0.022 |
| Mogcn | 0.965 ± 0.021 | 0.964 ± 0.019 | 0.951 ± 0.020 | 0.820 ± 0.049 | 0.821 ± 0.046 | 0.825 ± 0.033 |
| CustOmics | 0.968 ± 0.010 | 0.968 ± 0.010 | 0.962 ± 0.020 | 0.747 ± 0.064 | 0.743 ± 0.066 | 0.740 ± 0.068 |
| AttentionMOI | 0.941 ± 0.018 | 0.941 ± 0.018 | 0.943 ± 0.025 | 0.747 ± 0.078 | 0.744 ± 0.081 | 0.744 ± 0.084 |
| MOGAT | 0.9640.019 | 0.961 ± 0.026 | 0.950 ± 0.028 | 0.832 ± 0.045 | 0.819 ± 0.038 | 0.823 ± 0.037 |
| MCgnn | **0.992±0.008** | **0.992±0.009** | **0.982±0.031** | **0.852 ± 0.055** | **0.851 ± 0.055** | **0.855 ± 0.053** |

classification through its utilization of multitask view cooperation, information and label aggregation, omics-specific learning, and integration of diverse omics data sources. Furthermore, when confronted with more challenging multi-class tasks such as BRCA subtype classification and renal cancer classification, MCgnn surpasses existing methods by achieving higher scores for weighted F1_weighted and BACC metrics; thus, showcasing its remarkable proficiency in handling multi-class omics data classification challenges. For the classification of kidney cancer types, distinct differences among chromophobe renal cell carcinoma (KICH), clear renal cell carcinoma (KIRC), and papillary renal cell carcinoma (KIRP) are evident in the omics data. This leads to consistently high performance across various evaluation metrics for all compared methods. Consequently, kidney cancer type classification represents a relatively straightforward task compared to other classification challenges and serves as a proof-of-concept experiment for multi-class applications.

The performance distribution of four MV-GNN-based multi-omics data integration methods across different datasets is depicted in Fig. 3. It is evident that MCgnn demonstrated higher classification accuracy on these three datasets examined. Additionally, MCgnn demonstrates lower variance on the BRCA and LGG datasets, indicating its enhanced stability and generalizability. This highlights the importance of inter-omics information transfer and integration in multi-omics data classification tasks. Moreover, MCgnn outperforms the three compared algorithms in terms of weighted F1 score and BACC, indicating its superior ability to classifying minority classes.

To comprehensively evaluate MCgnn's generalization capabilities and assess for potential overfitting or underfitting, we compared the model's performance on both training and test sets. Table 4 presents the average Accuracy (Acc), weighted F1-score (F1_weighted), and Balanced Accuracy (BACC) for MCgnn across the four TCGA datasets (BRCA, LGG, KIPAN, and ROSMAP). Each metric is reported with its corresponding standard deviation over five-fold cross-validation.

As shown in Table 4, MCgnn consistently achieves high performance on both the training and test sets for all datasets. Notably, the performance gap between the training and test sets is minimal, indicating strong generalization and a successful avoidance of significant overfitting. For instance, on the BRCA dataset, the test accuracy is 0.791 ± 0.021, while the training accuracy is 0.881 ± 0.004. Similarly, for KIPAN, which is a relatively easier classification task, the model demonstrates near-perfect performance on both training and test sets (0.992 ± 0.008 test Acc vs. 1.000 ± 0.001 train Acc). This robust performance across both sets, coupled with low standard deviations, underscores the stability and reliability of MCgnn in handling complex multi-omics data. The intrinsic design of our model, including the use of dropout layers and the two-stage training strategy (pre-training followed by main training), significantly contributes to this effective mitigation of overfitting.

### 4.4. Performance of MCgnn across different types of omics data

In this section, MCgnn for the classification of three types of omics data (mRNA, DNA methylation, miRNA) was employed to compare its performance with methods utilizing only two types of omics data. The experimental setups encompassed pairwise combinations of any two omics data as well as the integration of all three omics data.

The results depicted in Fig. 4 of the bar chart demonstrate that integrating all three omics data into MCgnn surpasses the classification

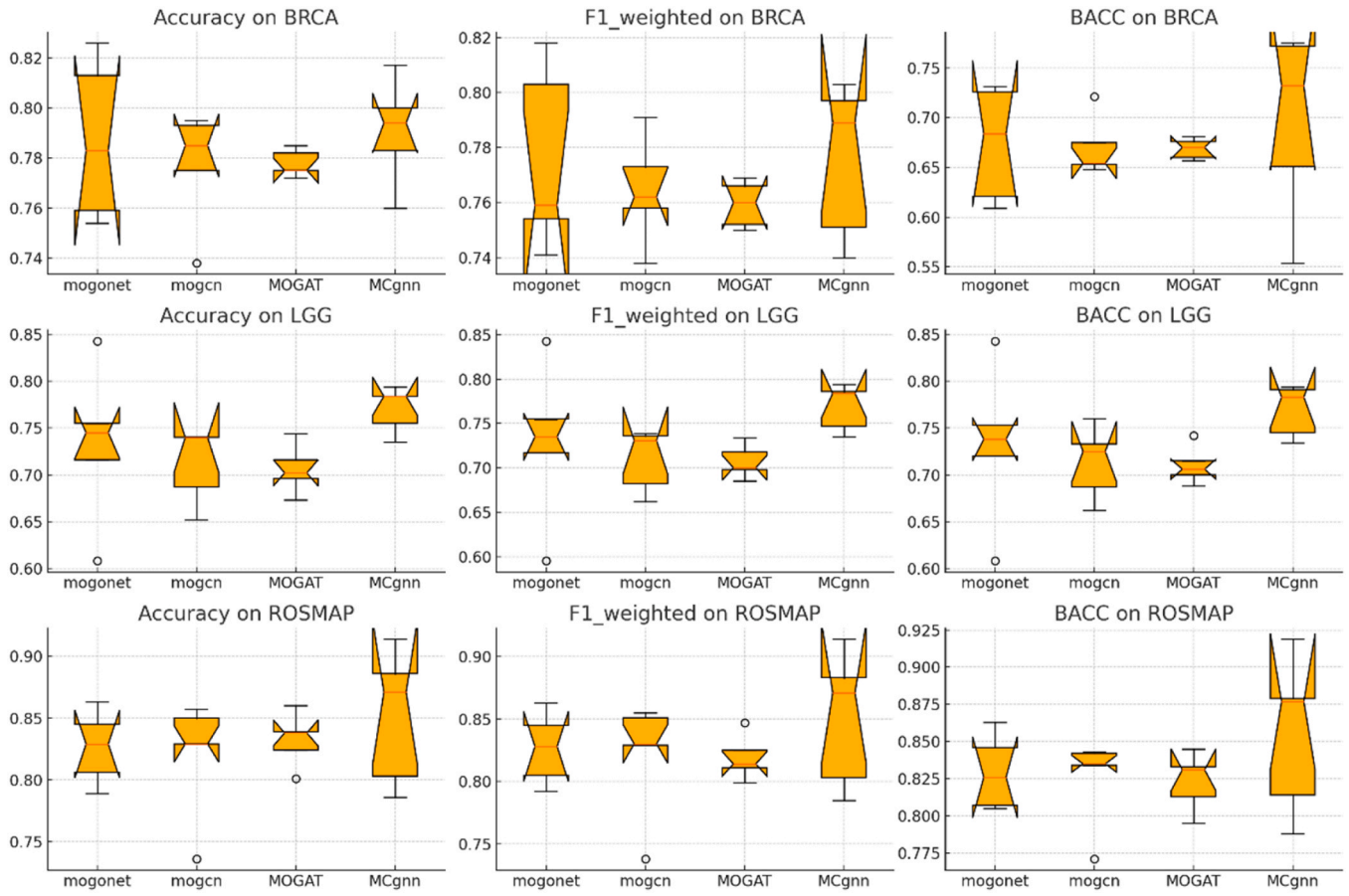



**Fig. 3.** Comparison of multi-omics data integration methods for BRCA, LGG, and ROSMAP datasets.

**Table 4**
Testing versus training figure of MCgnn on four datasets.

| Datasets | Test | | | Train | | |
|---|---|---|---|---|---|---|
| | Acc | F1_weighted | BACC | Acc | F1_weighted | BACC |
| BRCA | 0.791 ± 0.021 | 0.791 ± 0.008 | 0.742 ± 0.012 | 0.881 ± 0.004 | 0.888 ± 0.005 | 0.901 ± 0.006 |
| LGG | 0.771 ± 0.025 | 0.769 ± 0.026 | 0.770 ± 0.028 | 0.871 ± 0.054 | 0.864 ± 0.061 | 0.877 ± 0.051 |
| KIPAN | 0.992 ± 0.008 | 0.992 ± 0.009 | 0.982 ± 0.031 | 1.000 ± 0.001 | 1.000 ± 0.001 | 1.000 ± 0.001 |
| ROSMAP | 0.852 ± 0.055 | 0.851 ± 0.055 | 0.855 ± 0.053 | 0.949 ± 0.011 | 0.948 ± 0.011 | 0.949 ± 0.010 |

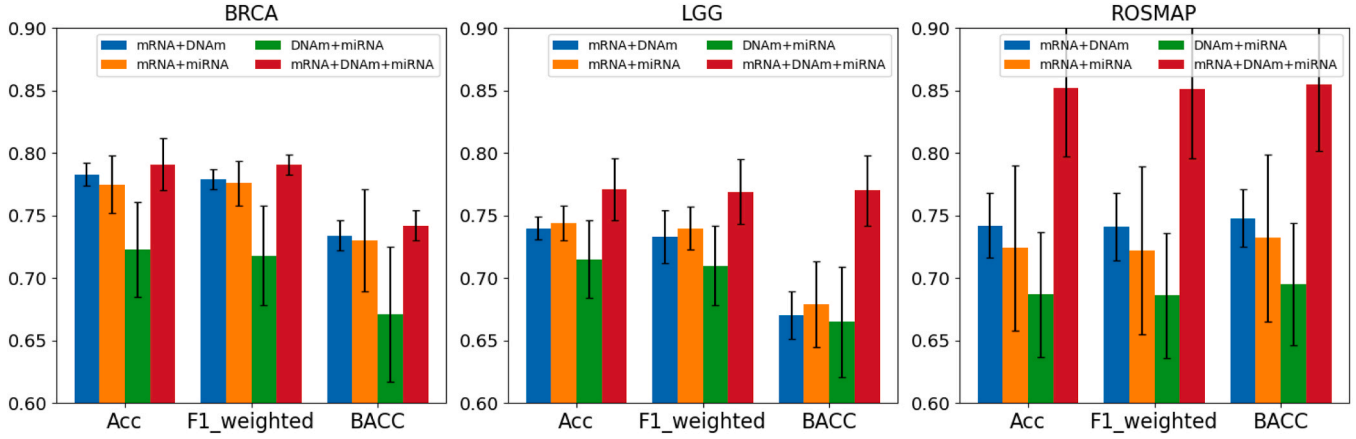



**Fig. 4.** Evaluation of MCgnn performance across diverse omics data types.

performance achieved using only two types of omics data. Notably, the incorporation of three omics data significantly enhances classification accuracy, highlighting its superior generalization capacity and stability. These findings underscore the effectiveness of a multi-view graph neural network in leveraging complementary information across diverse omics data, thereby capturing a more comprehensive range of biological features and ultimately improving classification outcomes. Conversely, employing fewer types of omics data leads to diminished performance due to information loss. Overall, these results validate the efficacy of MCgnn for integrating multi-omics data, particularly when incorporating heterogeneous types.

### 4.5. Ablation experiment

In this section, we analyze and compare the effectiveness of multi-view strategies and graph neural network components through ablation experiments to validate their contribution to the model's performance.

#### 4.5.1. Multi-view evaluation

The efficacy of MCgnn was evaluated in this section by comparing it with traditional concatenation techniques for omics data integration. In the conventional concatenation approach, all types of omics datasets are merged into a single large view. In contrast, in MCgnn each type of omics data is treated as a separate view and integrated using an MV-GNN. To ensure fairness, we compared MCgnn with a Single-view-Merged graph neural network (SMgnn), which employs techniques identical to MCgnn for graph construction and convolution. The experimental outcomes are depicted in Fig. 5.

The bar chart presented in Fig. 5 illustrates the comparative analysis of accuracy between MCgnn and SMgnn across the three TCGA datasets. The results demonstrate that MCgnn outperforms the large-view concatenation method in terms of both classification accuracy and stability. Specifically, across all datasets, MCgnn consistently exhibits significantly higher classification accuracy compared to traditional concatenation methods. Moreover, on the BRCA and LGG datasets, MCgnn manifests lower variance, indicating enhanced generalization capabilities and stability for these specific datasets.

These results validate the efficacy of the multi-view strategy, which preserves the inherent distinct structural information inherent to each omics data type, thereby facilitating a more comprehensive capture of complementary information between different omics layers and ultimately enhancing classification performance. In contrast, simplistic concatenation methods may compromise specific information associated with certain omics data types, potentially leading to diminished classification performance. Overall, our experimental findings indicate that the employment of a multi-view graph neural network in MCgnn represents an effective methodology for classifying multi-omics data, yielding exceptional performance across diverse datasets.

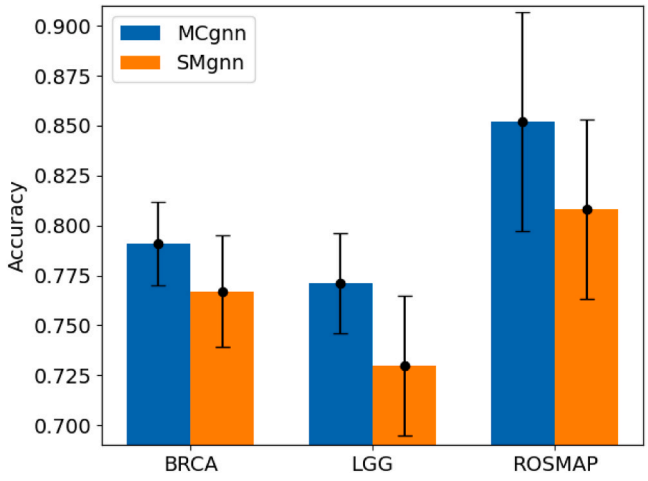



**Fig. 5.** Comparison of MCgnn and SMgnn on BRCA, LGG, and ROSMAP datasets.

#### 4.5.2. Graph neural network components evaluation

Specifically, the proposed methodology consists of three crucial components: graph convolution with adjacency matrix construction, inter-view attention mechanisms, and VCDN multi-view integration. To assess the significance of these components, the following ablation experiments were performed:

**MCgnn_A**: The graph convolution in the construction of the adjacency matrix is substituted with fully connected layers, while the subsequent components remain unchanged.

**MCgnn_B**: The inter-view attention mechanism is replaced by an equivalent number of fully connected layers.

**MCgnn_C:** The VCDN multi-view integration is replaced with an equivalent number of fully connected layers.

**MCgnn_D**: The Mahalanobis distance is replaced by Euclidean distance in graph neural network construction

This systematic approach enables us to methodically isolate and evaluate the impact of each component on the overall efficacy of the MCgnn model.

The box plot in Fig. 6 illustrates a comparison analysis of the accuracy of MCgnn and its variants across three TCGA datasets, showing the distribution of the results, the central tendency, the degree of dispersion, and the presence of outliers. The presented experimental results unequivocally demonstrate the superior performance of the original MCgnn method compared to alternative configurations. In this context, substituting the graph convolution component with fully connected layers leads to a noticeable decline in classification accuracy, highlighting the effectiveness of the density-based adjacency matrix in capturing local structural information among samples. Similarly, replacing both the inter-view attention component and the VCDN part with fully connected layers also results in a decrease in model performance, highlighting the critical role played by the inter-view attention mechanism and VCDN multi-view integration in extracting and fusing features from multi-omics data. Compared with MCgnn_D, the superiority of Mahalanobis distance as a metric for multi-omics data analysis is evident.

The efficacy and necessity of each component in the proposed methodology are validated by these findings, demonstrating the pivotal role of these design choices in enhancing the classification capability of MCgnn. The empirical evidence strongly supports our hypothesis that integrating these specific components significantly contributes to the robustness and accuracy of the MCgnn framework when dealing with complex multi-omics datasets.

### 4.6. Hyperparameter evaluation

An essential hyperparameter in MCgnn is $\boldsymbol{k}$, which adaptively determines the scaling factor for the density threshold used to construct the sample similarity network for the graph neural network structure. Capturing interactions between samples can significantly enhance information aggregation capabilities. However, setting a small value for $\boldsymbol{k}$ may lead to an excessively sparse similarity network, potentially overlooking crucial sample interactions. Conversely, when $\boldsymbol{k}$ is excessively large, the network becomes excessively compact and might include superfluous correlations or duplications among samples. Therefore, selecting an appropriate value for $\boldsymbol{k}$ is crucial to ensure optimal performance of MCgnn. Nonetheless, determining this optimal value depends on the topological structure of each dataset as it varies across datasets. To further demonstrate the impact of hyperparameter $\boldsymbol{k}$ on MCgnn's classification performance, MCgnn was trained using different values of $\boldsymbol{k}$ on BRCA, LGG, and ROSMAP datasets. As illustrated in Fig. 7, varying hyperparameter $\boldsymbol{k}$ does indeed affect MCgnn's classification

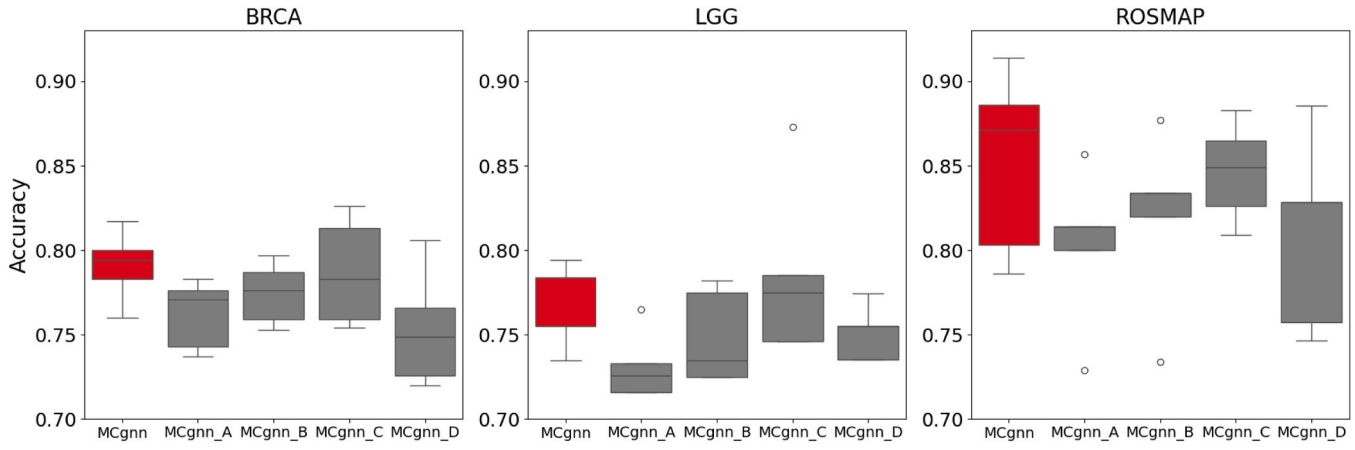



**Fig. 6.** Comparison of MCgnn and its variants on BRCA, LGG, and ROSMAP datasets.

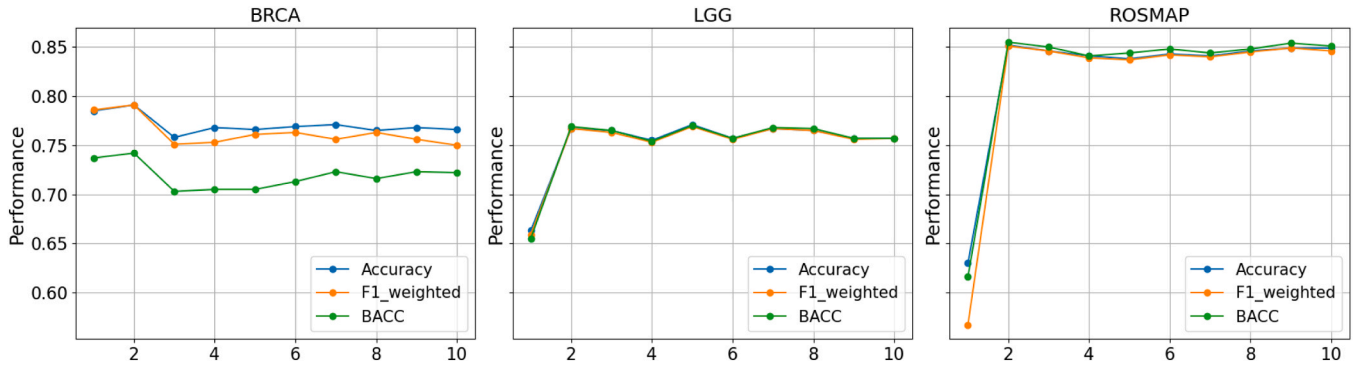



**Fig. 7.** Performance of MCgnn under different hyperparameters ***k***. on BRCA, LGG, and ROSMAP datasets.

performance with observed fluctuations as ***k*** changes; however, even with these variations in ***k*** values greater than 1, MCgnn consistently outperforms most existing methods.

### *4.7. Relevant biomarkers of significance identified*

Beyond achieving high classification performance, an important aspect of advanced predictive models in bioinformatics is their interpretability. MCgnn enhances the interpretability of cancer subtype classification by enabling the identification of significant biomarkers that directly contribute to its predictions. This provides valuable insights into the underlying biological mechanisms driving different cancer subtypes, thereby offering a more actionable reference for precision medicine. We employed a feature importance assessment strategy, which involves masking each feature and quantifying the resulting decline in model performance; higher importance scores indicate a greater contribution of the respective feature to the model's classification decision. By aggregating and ranking the importance across all views, we identified the top-n most significant features.

Feature importance is calculated by masking each feature and assessing the resulting decline in model performance. Higher importance scores indicate a greater contribution of the respective feature to the model. Aggregating and ranking the importance across all views allows for the identification of the top-n most significant features. The top 30 biomarkers with statistical importance for the BRCA and LGG datasets are presented in Tables 4 and 5, respectively. Notably, the identified biomarkers by MCgnn exhibited diverse functionalities and were associated with enriched biological processes specific to each disease. Detailed discussions on the results obtained from the dataset are provided in subsequent sections.

To assess the biological importance of the most highly ranked genes identified by MCgnn in terms of both mRNA expression and DNA methylation characteristics, we conducted gene set functional enrichment analysis utilizing the ToppGene Suite (Chen et al., 2009), a comprehensive tool that identifies significantly enriched biological annotations such as Gene Ontology (GO) (Gene Ontology Consortium, 2004) terms within a given gene set.

#### *4.7.1. Biomarkers related to breast cancer*

MCgnn identified 10 mRNA features, 15 DNA methylation features, and 5 miRNA features as the top 30 key biomarkers in the BRCA PAM50 subtype classification task, as shown in Table 5.

Several GO terms associated with the metabolism of folic acid, including folic acid itself, showed significant enrichment among the genes identified based on mRNA expression features: proton symporter activity (GO:0140211, p = 4.018E-4) and folic acid transmembrane transporter activity (GO:0008517, 2.408E-3). For example, refraining from folic acid or employing antifolate treatment could be an innovative precision-medicine approach for TNBCs characterized by metabolic inflexibility caused by heightened mitochondrial dysfunction (Coleman et al., 2021). Similarly, Raja et al (Rajeshkumar et al., 2024). discovered that promising breast cancer treatment involves the utilization of folic acid-conjugated clutenin nanoparticles. Additionally, molecular functions related to guanine were significantly enriched, including guanine transmembrane transporter activity (GO:0015208, p = 1.606E-3) and guanine transmembrane transport (GO:1903716, p = 1.549E-3). A breakthrough in breast cancer treatment involves a highly effective nanostructured liposome, functionalized with Larotaxel and adorned with a guanine-rich quadruplex nucleotide-lipid derivative, demonstrating remarkable efficacy against resistant breast cancer (Li et al.,

**Table 5**
Significant omics biomarkers identified within the BRCA dataset by MCgnn.

| Biomarker | Human Symbol | Gene ID | Omics Data Type |
|---|---|---|---|
| hsa-mir−1180 | | | miRNA expression |
| EDA2R | ectodysplasin A2 receptor | 60401 | mRNA expression |
| CHODL | chondrolectin | 140578 | mRNA expression |
| MPP6 | | | mRNA expression |
| hsa-mir−31 | | | miRNA expression |
| AGR3 | anterior gradient 3, protein disulphide isomerase family member | 155465 | mRNA expression |
| SLC43A3 | solute carrier family 43 member 3 | 29015 | mRNA expression |
| hsa-mir−496 | | | miRNA expression |
| WNT2 | Wnt family member 2 | 7472 | DNA methylation |
| hsa-mir−629 | | | miRNA expression |
| ARHGAP28 | Rho GTPase activating protein 28 | 79822 | DNA methylation |
| C5AR1 | complement C5a receptor 1 | 728 | DNA methylation |
| ELAC1 | elaC ribonuclease Z 1 | 55520 | DNA methylation |
| NEXN | nexilin F-actin binding protein | 91624 | DNA methylation |
| ULBP1 | UL16 binding protein 1 | 80329 | mRNA expression |
| SLC46A1 | solute carrier family 46 member 1 | 113235 | mRNA expression |
| ADCY4 | adenylate cyclase 4 | 196883 | DNA methylation |
| LAMA3 | laminin subunit alpha 3 | 3909 | DNA methylation |
| NFIL3 | nuclear factor, interleukin 3 regulated | 4783 | DNA methylation |
| hsa-mir−28 | | | miRNA expression |
| C10orf90 | chromosome 10 open reading frame 90 | 118611 | mRNA expression |
| GMDS | GDP-mannose 4,6-dehydratase | 2762 | DNA methylation |
| LCTL | lactase like | 197021 | mRNA expression |
| LTC4S | leukotriene C4 synthase | 4056 | DNA methylation |
| C22orf31 | chromosome 22 open reading frame 31 | 25770 | DNA methylation |
| GPR81 | | | mRNA expression |
| LOC153684 | | | DNA methylation |
| SAMD12 | sterile alpha motif domain containing 12 | 401474 | DNA methylation |
| LOC728640 | | | DNA methylation |
| FOXG1 | forkhead box G1 | 2290 | DNA methylation |

2021).

For DNA methylation, three GO terms associated with glutathione exhibited significant enrichment, including glutathione binding (GO:0043295, p = 1.080E-2), glutathione peroxidase activity (GO:0004602, p = 1.734E-2) and glutathione transferase activity (GO:0004364, p = 1.734E-2). PROteolysis TArgeting Chimeras (PROTACs) that specifically target ERα (estrogen receptor-α) have shown great potential as a new and promising approach in treating breast cancer (Zhou et al., 2023). Beatty et al (Beatty et al., 2018). discovered that restricting the accessibility of glutathione precursors amplifies the depletion of glutathione and enhances the efficacy of γ-glutamylcysteine ligase inhibitors in inducing cell death in TNBC cells during laboratory experiments.

#### 4.7.2. *Biomarkers related to low-grade glioma*

In the Low-Grade Glioma subtype classification task, MCgnn identified 17 mRNA features, 11 DNA methylation features, and 2 miRNA features as the top 30 key biomarkers, as shown in Table 6.

Relevant biological processes associated with the production of elastin were found to be significantly enriched among the genes

**Table 6**
Significant biomarkers identified within the LGG dataset by MCgnn.

| Biomarker | Human Symbol | Gene ID | Omics Data Type |
|---|---|---|---|
| NKX2–4 | NK2 homeobox 4 | 644524 | DNA methylation |
| ZNF702P | zinc finger protein 702, pseudogene | 79986 | DNA methylation |
| MIR663 | | | DNA methylation |
| SVEP1 | sushi, von Willebrand factor type A, EGF and pentraxin domain containing 1 | 79987 | mRNA expression |
| PRKCQ | protein kinase C theta | 5588 | mRNA expression |
| SPIC | Spi-C transcription factor | 121599 | DNA methylation |
| SLC16A10 | solute carrier family 16 member 10 | 117247 | mRNA expression |
| NTSR2 | neurotensin receptor 2 | 23620 | mRNA expression |
| OR10T2 | olfactory receptor family 10 subfamily T member 2 | 128360 | DNA methylation |
| FCER1A | Fc epsilon receptor Ia | 2205 | DNA methylation |
| CD3D | CD3 delta subunit of T-cell receptor complex | 915 | DNA methylation |
| TPSB2 | tryptase beta 2 | 64499 | DNA methylation |
| TPM3 | tropomyosin 3 | 7170 | DNA methylation |
| hsa-mir−891b | | | miRNA expression |
| XIRP2 | xin actin binding repeat containing 2 | 129446 | mRNA expression |
| VEPH1 | ventricular zone expressed PH domain containing 1 | 79674 | mRNA expression |
| LBX2 | ladybird homeobox 2 | 85474 | mRNA expression |
| PCBP3 | poly(rC) binding protein 3 | 54039 | mRNA expression |
| KRTAP4–8 | keratin associated protein 4–8 | 728224 | DNA methylation |
| BGN | biglycan | 633 | mRNA expression |
| PCSK6 | proprotein convertase subtilisin/kexin type 6 | 5046 | mRNA expression |
| C4orf19 | chromosome 4 open reading frame 19 | 55286 | mRNA expression |
| CA10 | carbonic anhydrase 10 | 56934 | mRNA expression |
| HIST1H3D | | | mRNA expression |
| SFT2D2 | SFT2 domain containing 2 | 375035 | mRNA expression |
| FMO2 | flavin containing dimethylaniline monoxygenase 2 | 2327 | mRNA expression |
| HBEGF | heparin binding EGF like growth factor | 1839 | mRNA expression |
| hsa-mir−320a | | | miRNA expression |
| OVCH2 | ovochymase 2 | 341277 | DNA methylation |
| LOC349196 | | | mRNA expression |

identified based on mRNA expression features, including: regulation of elastin biosynthetic process (GO:0051543, p = 7.264E-4), negative regulation of elastin biosynthetic process (GO:0051545, p = 7.264E-4) and elastin biosynthetic process (GO:0051542, p = 1.452E-3). Hinek et al (Hinek et al., 1999). discovered that the presence of suramin triggers an increased accumulation of EBP (elastin-binding protein) molecules on the surface of glioma cells, thereby enhancing the signals derived from elastin and promoting their advancement throughout the cell cycle. In addition, there was a significant enrichment of GO terms associated with T helper cells, including: T-helper 2 cell activation (GO:0035712, p = 1.452E-3), positive regulation of T-helper 2 cell activation (GO:2000570, p = 1.452E-3) and regulation of T-helper 2 cell activation (GO:2000569, p = 1.452E-3). With the progress in immunogenomics, it has been revealed that a significant proportion of tumors-specific antigens are recognizedd by T helper cells (THCs) (Zhao et al., 2023). Zheng et al (Zheng et al., 2016). propose that inducing specific T helper cells can inhibit the growth of experimental gliomas, which may be a new therapeutic approach for treating gliomas.

As for DNA methylation, there was a significant enrichment of three molecular functions associated with IgE, including high-affinity IgE receptor activity (GO:0019768, p = 5.022E-4), IgE receptor activity (GO:0019767, p = 3.010E-3) and IgE binding (GO:0019863, p = 3.511E-3). In the multivariate analysis of immune factors at baseline, a negative correlation was observed between glioma risk and elevated levels of respiratory allergen-specific IgE (odds ratio for allergen-specific IgE > 0.35 PAU/L: 0.59, p = .03) (Muskens et al., 2019). Furthermore, there is an enrichment of biological processes associated with interleukin, such as the facilitation of interleukin-3 production (GO:0032752, p = 8.715E-4), regulation of interleukin-3 production (GO:0032672, p = 1.307E-3), and the generation of interleukin-3 (GO:0032632, p = 1.742E-3). Interleukin-10 (IL-10) is a cytokine that exhibits dual functionality in various tumor types and contributes to the progression of glioma through the upregulation of KPNA2 (Zhang et al., 2019).

## 5. Conclusion

The rapid advancement of high-throughput technologies has made the integration and analysis of multi-omics data a powerful tool for unraveling the complexities of cancer. In response, we have developed MCgnn, a Multiview-Cooperated Graph Neural Network for Cancer Subtype Classification. MCgnn constructs similarity networks and employs convolution and inter-view attention mechanisms to facilitate collaborative exploration across multiple views, leveraging local structural information and exploiting complementary insights from different perspectives. Furthermore, MCgnn uses multi-task learning to seamlessly integrate information across tasks, achieving exceptional performance in cancer subtype classification. Notably, MCgnn also exhibits robustness, generalizability, and interpretability with immense potential in identifying significant biomarkers that can guide precision medicine.

Our future research will focus on integrating non-aligned multi-omics datasets, including those with missing omics samples, with the aim of enhancing the accuracy of model predictions by leveraging the valuable information encapsulated within these datasets. Additionally, we intend to investigate the prediction of drug responses in cell lines by combining cancer drug data. This research direction promises to enhance our understanding of disease causes and advance precision medicine development.

## CRediT authorship contribution statement

**Ming Jin:** Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Mingzhu Lou:** Resources, Project administration. **Shaobo Deng:** Validation, Software, Methodology. **Min Li:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Lei Wang:** Visualization, Validation, Investigation. **Hua Rao:** Visualization, Validation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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